

APM1220: Applied Multivariate Data Analysis

FA 2. Sample Geometry and Random Sampling

Problem 3.1 (8 points)

Given the data matrix

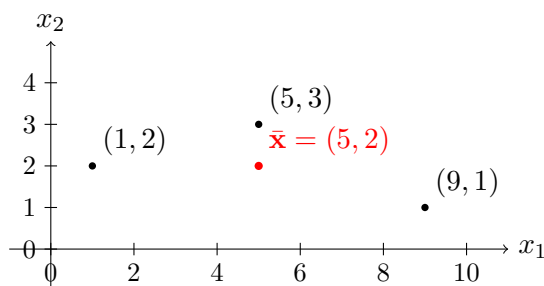
$$\mathbf{X} = \begin{pmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{pmatrix}, \quad n = 3, p = 2.$$

The column means are

$$\bar{x}_1 = \frac{9+5+1}{3} = 5, \quad \bar{x}_2 = \frac{1+3+2}{3} = 2, \quad \bar{\mathbf{x}}' = (5, 2).$$

(a) Scatter plot in $p = 2$ dimensions

Solution.



The three observations (9, 1), (5, 3), (1, 2) are plotted in the x_1 - x_2 plane, along with the sample mean point $\bar{\mathbf{x}} = (5, 2)$, which lies at the centroid of the three points.

(b) n -dimensional representation and deviation vectors

Solution. Each *variable* (column of $\mathbf{X}$) is now viewed as a vector in $n = 3$ -dimensional space:

$$\mathbf{y}_1 = \begin{pmatrix} 9 \\ 5 \\ 1 \end{pmatrix}, \quad \mathbf{y}_2 = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}.$$

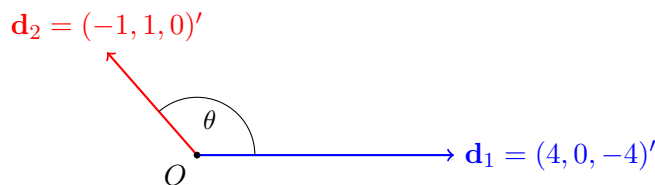
The deviation vectors are obtained by subtracting the corresponding sample mean (times $\mathbf{1}$) from each column:

$$\begin{aligned} \mathbf{y}_1 - \bar{x}_1 \mathbf{1} &= \begin{pmatrix} 9 \\ 5 \\ 1 \end{pmatrix} - 5 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ -4 \end{pmatrix} =: \mathbf{d}_1, \\ \mathbf{y}_2 - \bar{x}_2 \mathbf{1} &= \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} =: \mathbf{d}_2. \end{aligned}$$

Conceptually, $\mathbf{y}_1$ and $\mathbf{y}_2$ are vectors from the origin to the points (9, 5, 1) and (1, 3, 2) in $\mathbb{R}^3$; the deviation vectors $\mathbf{d}_1 = \mathbf{y}_1 - \bar{x}_1 \mathbf{1}$ and $\mathbf{d}_2 = \mathbf{y}_2 - \bar{x}_2 \mathbf{1}$ are the components of $\mathbf{y}_1, \mathbf{y}_2$ that are *perpendicular* to the equiangular line $\mathbf{1} = (1, 1, 1)'$ (since each $\mathbf{d}_i$ sums to zero).

(c) Deviation vectors from the origin: lengths, angle, and connection to $\mathbf{S}_n$ and $\mathbf{R}$

Solution.



(Sketch shown in a convenient 2-D projection since both vectors, together with the origin, span a plane; the true angle between them in $\mathbb{R}^3$ is computed below.)

Lengths:

$$L_1 = |\mathbf{d}_1| = \sqrt{4^2 + 0^2 + (-4)^2} = \sqrt{32} = 4\sqrt{2}, \quad L_2 = |\mathbf{d}_2| = \sqrt{(-1)^2 + 1^2 + 0^2} = \sqrt{2}.$$

Cosine of the angle between $\mathbf{d}_1$ and $\mathbf{d}_2$:

$$\mathbf{d}_1' \mathbf{d}_2 = (4)(-1) + (0)(1) + (-4)(0) = -4,$$

$$\cos \theta = \frac{\mathbf{d}_1' \mathbf{d}_2}{L_1 L_2} = \frac{-4}{(4\sqrt{2})(\sqrt{2})} = \frac{-4}{8} = -\frac{1}{2} \quad \implies \quad \theta = 120^\circ.$$

Connection to $\mathbf{S}_n$ and $\mathbf{R}$. With the divisor- n covariance matrix $\mathbf{S}_n$ ($s_{ii} = \mathbf{d}_i' \mathbf{d}_i / n$, $s_{ij} = \mathbf{d}_i' \mathbf{d}_j / n$):

$$s_{11} = \frac{L_1^2}{n} = \frac{32}{3}, \quad s_{22} = \frac{L_2^2}{n} = \frac{2}{3}, \quad s_{12} = \frac{\mathbf{d}_1' \mathbf{d}_2}{n} = \frac{-4}{3}.$$

The sample correlation coefficient is

$$r_{12} = \frac{s_{12}}{\sqrt{s_{11}s_{22}}} = \frac{-4/3}{\sqrt{(32/3)(2/3)}} = \frac{-4/3}{8/3} = -\frac{1}{2} = \cos \theta.$$

This confirms the general geometric fact: *the cosine of the angle between the two deviation vectors equals the sample correlation coefficient r_{12} between the two variables*, and the squared lengths of the deviation vectors are proportional ($\times n$) to the sample variances s_{11} and s_{22} .

Problem 3.6 (8 points)

$$\mathbf{X} = \begin{pmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{pmatrix}, \quad n = 3, p = 3.$$

(a) Matrix of deviations $\mathbf{X} - \mathbf{1}\bar{\mathbf{x}}'$ and its rank**Solution.** Column means:

$$\bar{x}_1 = \frac{-1 + 2 + 5}{3} = 2, \quad \bar{x}_2 = \frac{3 + 4 + 2}{3} = 3, \quad \bar{x}_3 = \frac{-2 + 2 + 3}{3} = 1, \quad \bar{\mathbf{x}}' = (2, 3, 1).$$

$$\mathbf{X} - \mathbf{1}\bar{\mathbf{x}}' = \begin{pmatrix} -1-2 & 3-3 & -2-1 \\ 2-2 & 4-3 & 2-1 \\ 5-2 & 2-3 & 3-1 \end{pmatrix} = \begin{pmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{pmatrix} =: \mathbf{D}.$$

Its determinant is

$$\det(\mathbf{D}) = -3(1 \cdot 2 - 1 \cdot (-1)) - 0(0 \cdot 2 - 1 \cdot 3) + (-3)(0 \cdot (-1) - 1 \cdot 3) = -3(3) - 0 + (-3)(-3) = -9 + 9 = 0.$$

Since $\det(\mathbf{D}) = 0$, $\mathbf{D}$ is **not of full rank** ($\text{rank} < 3$). This is expected: because each column of a mean-corrected data matrix sums to zero, every column of $\mathbf{D}$ lies in the $(n-1)$ -dimensional subspace of $\mathbb{R}^n$ orthogonal to $\mathbf{1}$. With $n = 3$, this subspace has dimension 2, so the $p = 3$ column vectors of $\mathbf{D}$ (each in $\mathbb{R}^3$) cannot be linearly independent — at most $\text{rank}(\mathbf{D}) = \min(p, n-1) = 2$.

(b) Sample covariance matrix $\mathbf{S}$ and generalized variance $|\mathbf{S}|$

Solution. $\mathbf{S} = \frac{1}{n-1} \mathbf{D}'\mathbf{D} = \frac{1}{2} \mathbf{D}'\mathbf{D}$. Writing the columns of $\mathbf{D}$ as $\mathbf{d}_1 = (-3, 0, 3)'$, $\mathbf{d}_2 = (0, 1, -1)'$, $\mathbf{d}_3 = (-3, 1, 2)'$:

$$\mathbf{D}'\mathbf{D} = \begin{pmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{pmatrix}, \quad \mathbf{S} = \frac{1}{2} \begin{pmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{pmatrix} = \begin{pmatrix} 9 & -\frac{3}{2} & \frac{15}{2} \\ -\frac{3}{2} & 1 & -\frac{1}{2} \\ \frac{15}{2} & -\frac{1}{2} & 7 \end{pmatrix}.$$

Because $\mathbf{D}$ has rank $2 < 3$, $\mathbf{D}'\mathbf{D}$ (and hence $\mathbf{S}$) is **singular**:

$$|\mathbf{S}| = 0.$$

Geometric interpretation. The generalized sample variance $|\mathbf{S}|$ is proportional to the squared volume of the parallelotope spanned by the $p = 3$ deviation vectors $\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3$ in n -space (specifically $|\mathbf{S}| = \text{vol}^2 / (n-1)^p$). Since $|\mathbf{S}| = 0$, this parallelotope has **zero volume**: the three deviation vectors are linearly dependent and therefore all lie in a common plane (a 2-dimensional subspace) rather than spanning a genuine 3-dimensional solid. Equivalently, the scatter of the $n = 3$ points does not "fill up" all $p = 3$ dimensions — one variable is redundant given the other two.

(c) Total sample variance

Solution. The total sample variance is the trace of $\mathbf{S}$ (sum of the sample variances of the individual variables):

$$\text{tr}(\mathbf{S}) = s_{11} + s_{22} + s_{33} = 9 + 1 + 7 = \boxed{17}.$$

Problem 3.9 (8 points)

$$\mathbf{X} = \begin{pmatrix} 12 & 17 & 29 \\ 18 & 20 & 38 \\ 14 & 16 & 30 \\ 20 & 18 & 38 \\ 16 & 19 & 35 \end{pmatrix}, \quad n = 5,$$

where $x_3 = x_1 + x_2$ (total of the two tests).

(a) Mean-corrected data matrix and linear dependence

Solution. Column means:

$$\bar{x}_1 = \frac{12 + 18 + 14 + 20 + 16}{5} = 16, \quad \bar{x}_2 = \frac{17 + 20 + 16 + 18 + 19}{5} = 18, \\ \bar{x}_3 = \frac{29 + 38 + 30 + 38 + 35}{5} = 34.$$

Note $\bar{x}_1 + \bar{x}_2 = 34 = \bar{x}_3$, as expected. The mean-corrected matrix is

$$\mathbf{X} - \mathbf{1}\bar{\mathbf{x}}' = \begin{pmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{pmatrix} =: \mathbf{D}.$$

Column 3 of $\mathbf{D}$ equals the sum of columns 1 and 2 in every row:

$$(-4) + (-1) = -5, \quad 2 + 2 = 4, \quad (-2) + (-2) = -4, \quad 4 + 0 = 4, \quad 0 + 1 = 1.$$

Hence the columns are linearly dependent, with

$$\mathbf{a}' = (a_1, a_2, a_3) = (1, 1, -1), \quad \text{since } a_1\mathbf{d}_1 + a_2\mathbf{d}_2 + a_3\mathbf{d}_3 = \mathbf{0}.$$

(b) Sample covariance matrix, $|\mathbf{S}| = 0$, and $\mathbf{S}\mathbf{a} = \mathbf{0}$

Solution. With $\mathbf{d}_1 = (-4, 2, -2, 4, 0)'$, $\mathbf{d}_2 = (-1, 2, -2, 0, 1)'$, $\mathbf{d}_3 = (-5, 4, -4, 4, 1)'$ and $n - 1 = 4$:

$$\mathbf{D}'\mathbf{D} = \begin{pmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{pmatrix}, \quad \mathbf{S} = \frac{1}{4}\mathbf{D}'\mathbf{D} = \begin{pmatrix} 10 & 3 & 13 \\ 3 & \frac{5}{2} & \frac{11}{2} \\ 13 & \frac{11}{2} & \frac{37}{2} \end{pmatrix}.$$

Because $\mathbf{d}_3 = \mathbf{d}_1 + \mathbf{d}_2$, the columns of $\mathbf{D}$ are linearly dependent ($\text{rank } \mathbf{D} \leq 2 < 3$), so $\mathbf{S}$ is singular and

$$|\mathbf{S}| = 0.$$

Verification that $\mathbf{S}\mathbf{a} = \mathbf{0}$ for $\mathbf{a} = (1, 1, -1)'$:

$$\mathbf{S}\mathbf{a} = \begin{pmatrix} 10(1) + 3(1) + 13(-1) \\ 3(1) + \frac{5}{2}(1) + \frac{11}{2}(-1) \\ 13(1) + \frac{11}{2}(1) + \frac{37}{2}(-1) \end{pmatrix} = \begin{pmatrix} 10 + 3 - 13 \\ 3 + 2.5 - 5.5 \\ 13 + 5.5 - 18.5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Since $\mathbf{S}\mathbf{a} = \mathbf{0} = 0 \cdot \mathbf{a}$, the vector $\mathbf{a} = (1, 1, -1)'$ (or any nonzero scalar multiple of it) is an eigenvector of $\mathbf{S}$ corresponding to the eigenvalue $\lambda = 0$.

(c) **Verifying** $x_3 = x_1 + x_2$

Solution. Checking each row of $\mathbf{X}$ directly:

$$12 + 17 = 29, \quad 18 + 20 = 38, \quad 14 + 16 = 30, \quad 20 + 18 = 38, \quad 16 + 19 = 35,$$

each of which matches the corresponding entry of column 3. Hence for every observation,

$$x_1 + x_2 - x_3 = 0 \quad \Longleftrightarrow \quad a_1x_1 + a_2x_2 + a_3x_3 = 0 \text{ with } a_1 = 1, a_2 = 1, a_3 = -1,$$

confirming the exact linear dependence among the three columns (as used in parts (a) and (b)).

Problem 3.18 (8 points)

Summary statistics for 2001 state energy consumption (x_1 = petroleum, x_2 = natural gas, x_3 = hydroelectric, x_4 = nuclear; quadrillion BTUs):

$$\bar{\mathbf{x}} = \begin{pmatrix} 0.766 \\ 0.508 \\ 0.438 \\ 0.161 \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 0.856 & 0.635 & 0.173 & 0.096 \\ 0.635 & 0.568 & 0.128 & 0.067 \\ 0.173 & 0.128 & 0.171 & 0.039 \\ 0.096 & 0.067 & 0.039 & 0.043 \end{pmatrix}.$$

(The off-diagonal entry $s_{23} = s_{32}$ is taken as 0.128, the symmetric value, to correct a printing rounding inconsistency in the source table.)

(a) Mean and variance of total energy consumption

Solution. Total consumption is the linear combination $T = x_1 + x_2 + x_3 + x_4 = \mathbf{c}'\mathbf{x}$ with $\mathbf{c} = (1, 1, 1, 1)'$.

Mean:

$$\bar{T} = \mathbf{c}'\bar{\mathbf{x}} = 0.766 + 0.508 + 0.438 + 0.161 = 1.873 \text{ (quadrillion BTUs)}.$$

Variance: $\text{Var}(T) = \mathbf{c}'\mathbf{S}\mathbf{c} = \sum_i \sum_j s_{ij}$, i.e., the sum of *all* entries of $\mathbf{S}$:

$$\begin{aligned} \text{Var}(T) &= (0.856 + 0.635 + 0.173 + 0.096) + (0.635 + 0.568 + 0.128 + 0.067) \\ &\quad + (0.173 + 0.128 + 0.171 + 0.039) + (0.096 + 0.067 + 0.039 + 0.043) \\ &= 1.760 + 1.398 + 0.511 + 0.245 \\ &= \boxed{3.914}. \end{aligned}$$

(b) Excess of petroleum over natural gas; covariance with total

Solution. Let $D = x_1 - x_2 = \mathbf{b}'\mathbf{x}$ with $\mathbf{b} = (1, -1, 0, 0)'$.

Mean:

$$\bar{D} = \bar{x}_1 - \bar{x}_2 = 0.766 - 0.508 = 0.258.$$

Variance:

$$\text{Var}(D) = \mathbf{b}'\mathbf{S}\mathbf{b} = s_{11} - 2s_{12} + s_{22} = 0.856 - 2(0.635) + 0.568 = 0.154.$$

Covariance of D with T (part a):

$$\text{Cov}(D, T) = \mathbf{b}'\mathbf{S}\mathbf{c} = (s_{11} + s_{12} + s_{13} + s_{14}) - (s_{21} + s_{22} + s_{23} + s_{24}).$$

Using the row sums already computed in part (a) (row 1 sum = 1.760, row 2 sum = 1.398):

$$\text{Cov}(D, T) = 1.760 - 1.398 = \boxed{0.362}.$$

Problem 3.20 (8 points)

Snow-removal data for $n = 25$ Wisconsin storms: x_1 = storm duration (hrs), x_2 = crew/machine hours to clear snow.

Summary sums (computed directly from Table 3.2):

$$\begin{aligned}\sum x_{j1} &= 235.5, & \sum x_{j2} &= 481.8, & \sum x_{j1}^2 &= 2557.75, \\ \sum x_{j2}^2 &= 10778.98, & \sum x_{j1}x_{j2} &= 4861.90.\end{aligned}$$

From these,

$$\begin{aligned}\bar{x}_1 &= \frac{235.5}{25} = 9.420, & \bar{x}_2 &= \frac{481.8}{25} = 19.272, \\ s_{11} &= \frac{\sum x_{j1}^2 - n\bar{x}_1^2}{n-1} = \frac{2557.75 - 25(9.42)^2}{24} = 14.1392, \\ s_{22} &= \frac{\sum x_{j2}^2 - n\bar{x}_2^2}{n-1} = \frac{10778.98 - 25(19.272)^2}{24} = 62.2388, \\ s_{12} &= \frac{\sum x_{j1}x_{j2} - n\bar{x}_1\bar{x}_2}{n-1} = \frac{4861.90 - 25(9.42)(19.272)}{24} = 13.4727.\end{aligned}$$

(a) Mean and variance of $x_2 - x_1$ from the summary statistics

Solution. Mean:

$$\overline{x_2 - x_1} = \bar{x}_2 - \bar{x}_1 = 19.272 - 9.420 = 9.852.$$

Variance:

$$\text{Var}(x_2 - x_1) = s_{11} + s_{22} - 2s_{12} = 14.1392 + 62.2388 - 2(13.4727) = \boxed{49.4326}.$$

(b) Mean and variance from the individual differences $d_j = x_{j2} - x_{j1}$

Solution. Computing $d_j = x_{j2} - x_{j1}$ for each $j = 1, \dots, 25$ and summing:

$$\sum_{j=1}^{25} d_j = 246.3, \quad \sum_{j=1}^{25} d_j^2 = 3612.93.$$

Mean:

$$\bar{d} = \frac{246.3}{25} = 9.852.$$

Variance:

$$s_d^2 = \frac{\sum d_j^2 - n\bar{d}^2}{n-1} = \frac{3612.93 - 25(9.852)^2}{24} = \frac{3612.93 - 2426.567}{24} = \frac{1186.363}{24} = \boxed{49.4326}.$$

Comparison. The mean (9.852) and variance (49.4326) obtained by first computing individual differences d_j and then their sample mean/variance are **identical** to those obtained in part (a) using the summary statistics $\bar{x}_1, \bar{x}_2, s_{11}, s_{22}, s_{12}$. This confirms the general algebraic identities

$$\overline{x_2 - x_1} = \bar{x}_2 - \bar{x}_1, \quad \text{Var}(x_2 - x_1) = s_{11} + s_{22} - 2s_{12},$$

which hold exactly for any bivariate sample, so the two computational routes must agree.